# SCHOOL OF ENGINEERING PROJECT
## WEEKLY PROGRESS REPORT

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Objectives of the Week :

1. Built a simple UI where users can upload CVs and job descriptions for the screening system.
2. Packaged the TalentLens app using Docker so it can run the same way on different machines without setup issues.

Objectives Achieved and Work Done :

This week, I deployed the TalentLens CV screening application that I built in the prior weeks. My first objective was to package the app with Docker so it can be run by any user that may not have Python or necessary libraries on their laptop. To achieve this, I created the following four files:

- Dockerfile: outlines the version of Python, installs the libraries, and then runs the Streamlit app
- requirements.txt: describes all libraries including streamlit, groq, faiss-cpu, and sentence-transformers
- docker-compose.yml: enables the app to be run with a command (docker compose up --build) and constructs a volume that caches the HuggingFace model to avoid downloading it each time
- .dockerignore: removes extraneous files and notebooks, resulting in a minimal Docker image

I also tried to deploy the app on the cloud using Render. I made a repository on GitHub, added my project files and then linked the repository to Render with Docker. The deployment was successful, but the app crashed continually because of an out of memory error. I found the cause was that sentence-transformers produced a model that was using RAM over Render's free tier limit of 512MB and was using about 400MB. I attempted a few modifications, but crash issues with the free tier couldn't be resolved.

Due to issues with cloud deployment, I ran the app locally with Docker Desktop. After I did the initial installation, I faced permission issues with Docker's settings and WSL. I used PowerShell to fix these issues and run my container.

Now that the app is running, I can see that the functions inside Docker is complete. Users can interact with the app to upload their CVs and job descriptions and run the screening system.

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| Supervisor's Comment :<br><br>_____<br>Date/Signature |